Tesselle

— GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Imagine it. Keep what matters. Get it made.

Project Tesselle is a personal commerce agent that turns any spark - words, a photo, a video, a glance through smart glasses - into a brief it protects all the way to delivery.

project-tesselle.web.app



PROJECT TESSELLE

TEAM TESSELLE · ASHFAQUE RIFAYE

Inspiration is easy. Getting the right thing is not.

≠

Similar isn’t suitable

A product that looks right can be 4 cm too deep, need drilling, or miss the one compartment that mattered.

?

Search misses aren’t answers

When stock doesn't fit, shoppers either give up or quietly buy a compromise - and nobody can make the thing they actually wanted.



Intent gets lost in the middle

Seller counteroffers, substitutions, quote changes and device switches rewrite requirements without anyone noticing.

The missing piece isn't more products. It's an agent that **preserves what the customer needs** through the messy commercial middle.

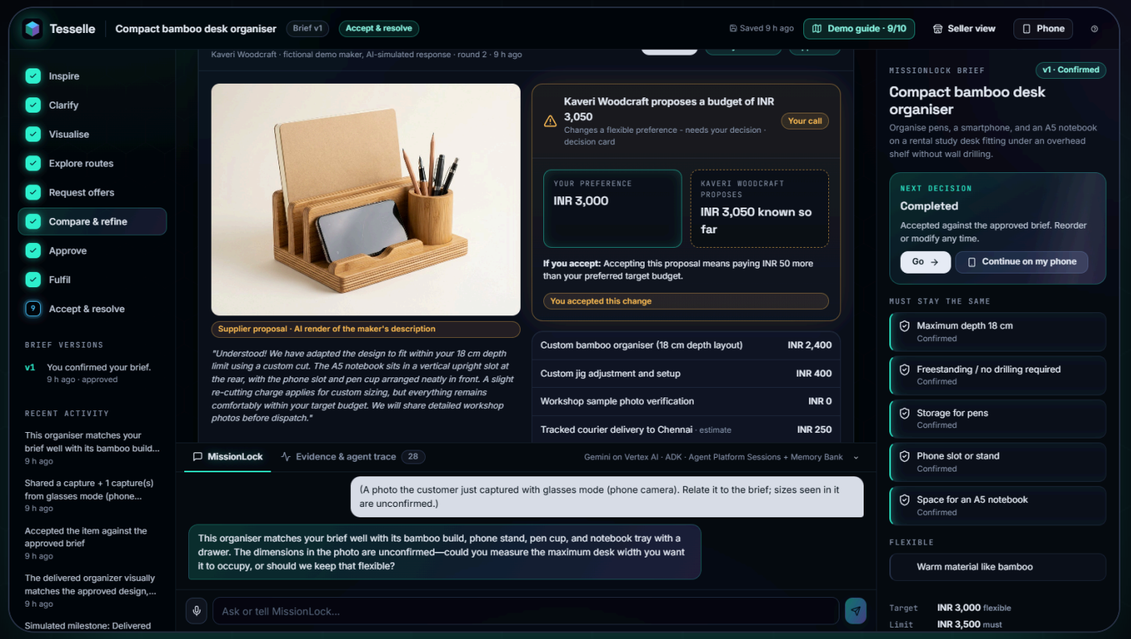
One agent. One brief. Four credible routes.

MissionLock - the intent safeguard

A living, versioned brief: must-haves, flexible preferences, budget, unknowns and acceptance checks. Nothing material changes without the customer's explicit decision.

Fulfilment - buy, source, customise or make

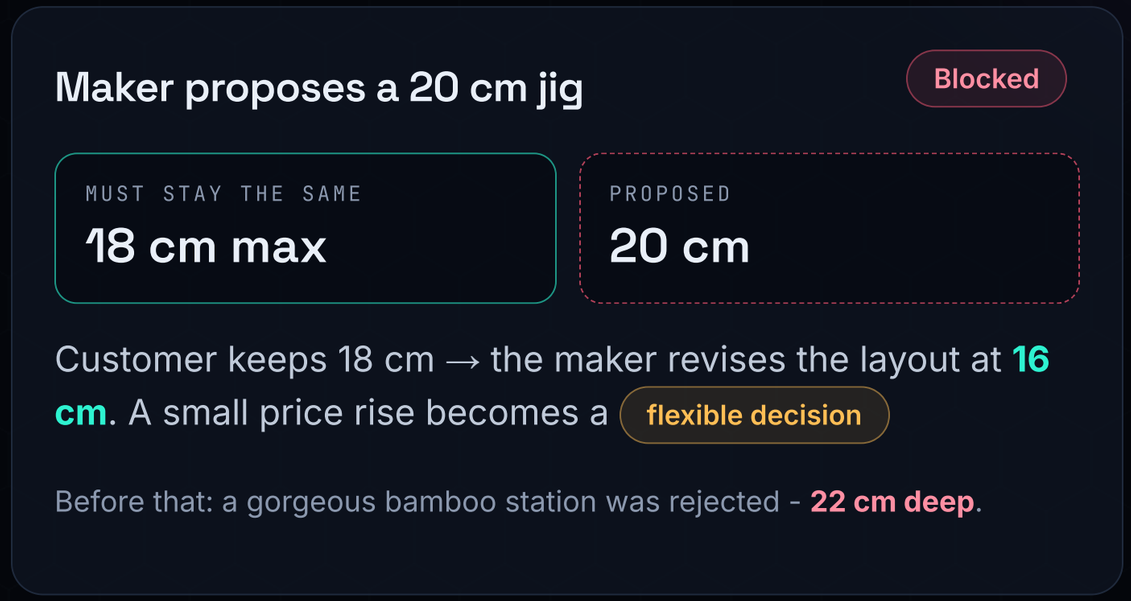
Every route is checked against the same brief, requirement by requirement. When nothing fits, a maker gets a clean brief - and their counteroffer becomes the customer's decision, never a silent swap.



Language models propose. Deterministic code decides.

- **Fit verdicts** - meets / possible with a change / not suitable, computed in code from structured requirements.
- **Quote deviations** - every offered spec is diffed against the brief version it answers.
- **Approval gates** - no stale quote, no changed brief, no unmet must-have, no duplicate order.
- **Honest uncertainty** - photo sizes are "Needs checking"; unknown charges are never folded into a total.

Gemini agents only *explain* the verdicts - a cheaper or prettier item that breaks a must-have is never promoted.

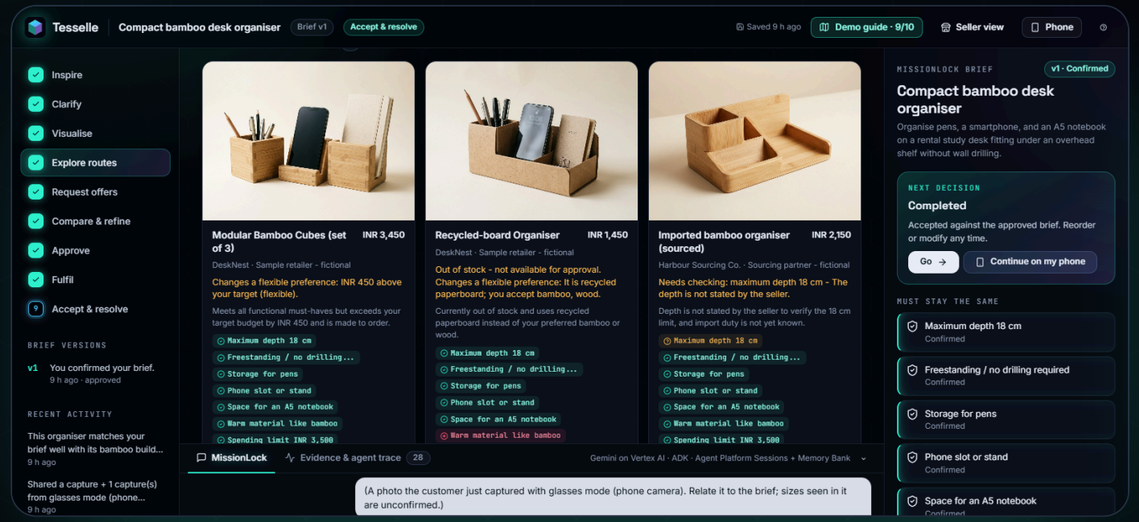


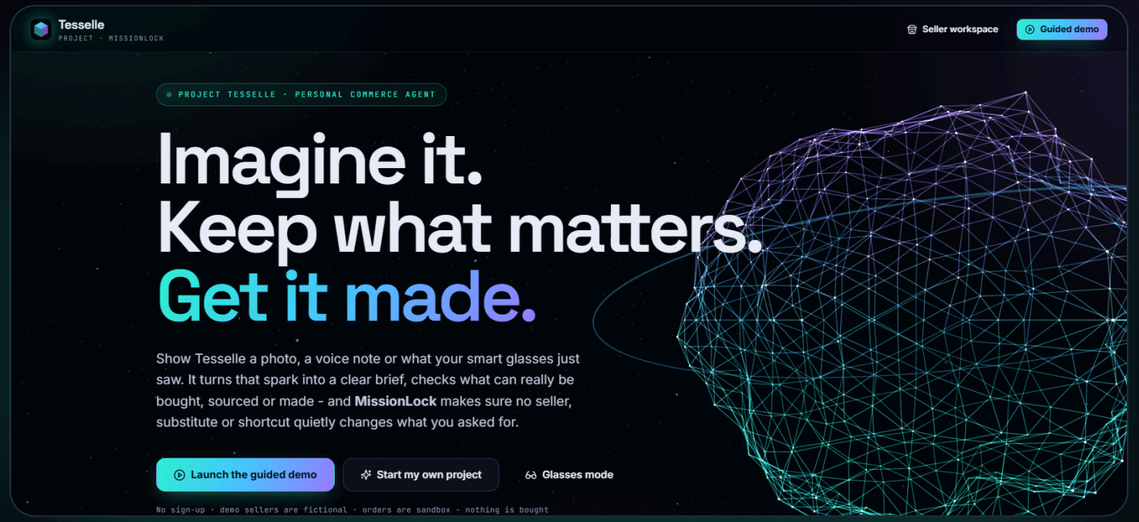
The full journey, live on Google Cloud.

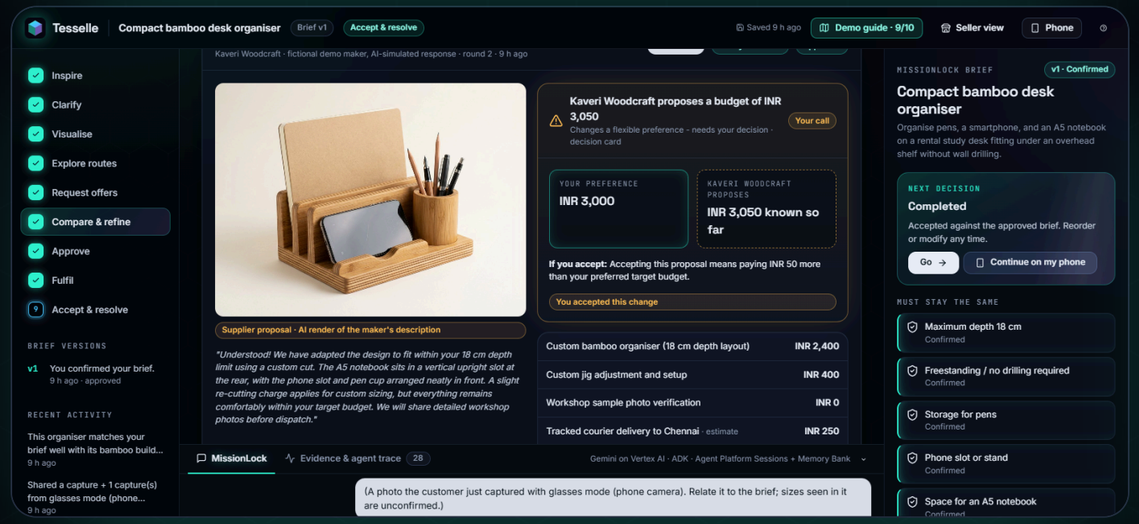
1. **Inspiration**
   Desk photo + "18 cm shelf, I rent, no drilling, about ₹3,000"
2. **Intent**
   Gemini drafts; the customer confirms must-haves vs flexible
3. **Concept**
   Labelled render; pin an area to refine
4. **Existing stock**
   22 cm beauty rejected - honest "no suitable match"
5. **Maker request**
   Privacy-filtered brief, reviewed before sending
6. **Counteroffer**
   20 cm proposal blocked by MissionLock
7. **Resolution**
   Revised 16 cm layout keeps the must-have
8. **Device handoff**
   QR → same pending decision on the phone, spoken recap
9. **Approval**
   Exact spec + quote + brief version; sandbox order
10. **Aftercare**
    Gemini vision checks delivery against approved checks

All ten beats are asserted end-to-end against the live deployment by tools/e2e_storyboard.py.

A calm, futuristic workspace - every decision explained.









Nine Gemini agents, one graph workflow, real memory.

Multimodal intake

ADK LlmAgent on Gemini 3.8 Flash with typed output: text, voice, photos, **video** and generated images.

Route explorer

ADK 2 graph Workflow: three scouts in parallel → JoinNode → analyst. Firestore vector search over Gemini embeddings.

Creative studio

Art-director agent + Gemini image generation; click-to-pin visual prompting; flags edits that would touch the brief.

Negotiation

AI-simulated maker agent quotes per brief version; deviation-explainer agent writes decision cards.

Memory & continuity

Agent Platform Sessions per project and **Memory Bank** for durable preferences ("I rent - no drilling").

Voice & vision

Concierge with function tools, Gemini TTS replies, recap agent, and a delivery inspector that knows what a photo can't prove.

Built end-to-end on Google Cloud.

SURFACES

React + Three.js workspace
Phone surface (QR handoff)
Glasses HUD capture
PWA share target (Ray-Ban Meta)
Seller workspace

Firebase Hosting CDN

CLOUD RUN · FASTAPI + GOOGLE ADK 2

9 LlmAgents with output schemas & tools
Graph Workflow (parallel scouts + join)
MissionLock guard - deterministic
NDJSON progress streams with heartbeats
OpenTelemetry → Cloud Trace
Least-privilege service account · no API keys

VERTEX AI / AGENT PLATFORM

Gemini 3.8 Flash (text, vision, video)
Gemini 3.1 Flash Image
Gemini Embedding 2
Gemini 2.5 Flash TTS
Agent Runtime: Sessions + Memory Bank

FIRESTORE

Projects, immutable brief versions, quotes, orders, event trace + vector index

CLOUD STORAGE

Private media: inspirations, concepts, proposals, evidence

CLOUD BUILD + ARTIFACT REGISTRY

Container builds, asia-southeast1

LOGGING · TRACE · BILLING LABELS

Every Gemini call labelled app=tesselle; daily cost guard

From a sentence to a glance through your glasses.

Speak it

Voice notes understood directly by Gemini.

Capture it

Live camera photos and short clips; video understanding.

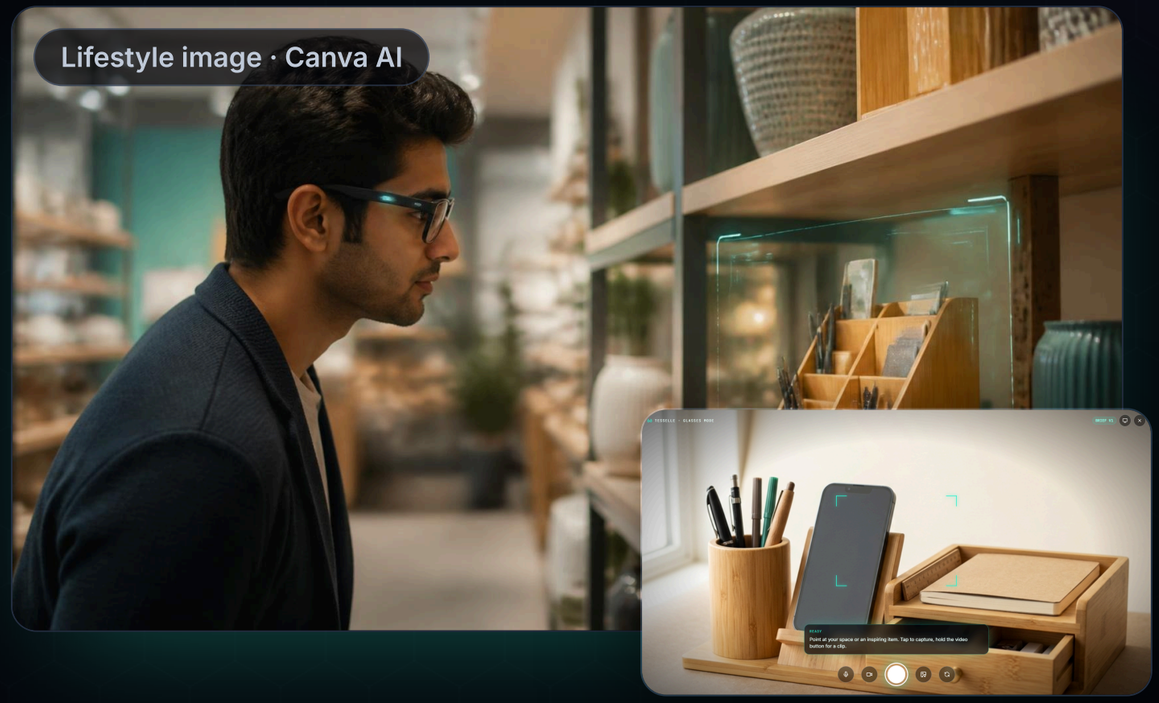
Imagine it

Gemini renders an inspiration, labelled as a concept.

Glance at it

Glasses HUD + PWA share target; replies spoken aloud.

Ray-Ban Meta captures sync to the phone and are shared straight to Tesselle. Direct camera streaming via Meta's native SDK is on the roadmap.



Unmet demand becomes actionable demand.

Customers

Less repeated searching and explaining, visible trade-offs, access to custom making - and the confidence that what arrives is what they approved.

Makers & retailers

Qualified, structured briefs instead of vague enquiries; a way to serve demand existing stock can't. They can decline or propose - never be misrepresented.

Model

Disclosed fee on completed sourcing or custom orders; seller subscriptions for brief and quote management; optional design and sampling services.

What we'll validate (not claimed as achieved)

Brief completion · request-to-quote conversion · suitable-quote acceptance · requirement changes after approval · delivered-item acceptance - against an ordinary search-and-enquiry journey, with consented user testing.

Honest by default. Safe by construction.

- Four unmistakable visual states: concept, existing product, supplier proposal, approved sample.
- Must-haves change only through a labelled decision that creates a new brief version.
- "No suitable match found in the sources checked" - an honest dead end beats a fabricated success.
- Demo sellers are fictional and AI-simulated; orders are sandbox - and labelled so.
- Sellers receive a privacy-filtered brief - no name, address or home imagery unless the customer opts in.
- The conversational agent cannot send requests, accept seller changes or approve spending.
- Vertex AI only, one service identity, no API keys; billing labels on every Gemini call.
- Public-demo cost guard caps daily model and image calls.

From one desk organiser to a demand engine.

NOW

Live prototype

Full storyboard, multimodal + glasses capture, memory, sandbox orders.

NEXT

Real makers

Onboard participating makers in Tamil Nadu and Karnataka; verified quote workflow.

THEN

Live voice & glasses

Gemini Live conversations; Meta wearables SDK for direct capture.

LATER

Shared demand

Opt-in pooling of compatible briefs when minimum quantities block a maker.

Tesselle

— TRY IT

Your intent, kept.

project-tesselle.web.app

tesselle-448771854429.asia-southeast1.run.app

Launch the guided demo - about five minutes, no sign-up. Demo sellers are fictional; orders are sandbox.

